Skills Summary

Reading Base-Ten Models: Hundreds, Tens, and Ones

Use this reference while completing your worksheet or when studying.

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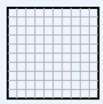
1. Reading a model (read-base-ten-model)

The three blocks:  = one **hundred** ·  = one **ten** · = one **one**

Rule: count flats, then rods, then units, and write one digit in each place: (flats) $\times$ 100 + (rods) $\times$ 10 + (units) $\times$ 1. An empty group still needs a digit — write 0 to hold its place.

Example: A model shows 3 flats, 5 rods, and 4 units. What number is it?
Step 1. Each kind of block has one job: flats count hundreds, rods count tens, units count ones — so I count each group and put its count in that place.
Step 2. $3 \times 100 = 300$, $5 \times 10 = 50$, $4 \times 1 = 4$, and $300 + 50 + 4 = \mathbf{354}$

Try it: A model shows 2 flats, no rods, and 7 units. What number is it?

check: **207**

2. Expanded form (expanded-form-from-model)

Expanded form writes what each group of blocks is worth, then adds:
$347 = 300 + 40 + 7$ (3 flats + 4 rods + 7 units).
The digit tells *how many*; the place tells *how much each one is worth.*

Example: Write 6 flats, 1 rod, and 8 units in expanded form.
Step 1. Expanded form is just the block count with each group's value written out, so I turn each group into its value before adding.
Step 2. 6 flats = 600, 1 rod = 10, 8 units = 8, so the expanded form is $600 + 10 + 8 = \mathbf{618}$

Try it: Write 4 flats, 5 rods, and 3 units in expanded form and as a number.

check: $400 + 50 + 3 = \mathbf{453}$

3. Trading rods for flats (regroup-tens-into-hundreds)

Rule: a place can only hold the digits 0–9. If a model shows ten or more rods, trade: 10 rods = 1 flat. Trading never changes the number, only how it is shown.

Example: A model shows 1 flat, 12 rods, and 4 units.
Step 1. Twelve rods will not fit in the tens place, so I trade ten of them for one flat before reading any digits.
Step 2. 12 rods = 1 flat +2 rods, so the model is 2 flats, 2 rods, 4 units: $200 + 20 + 4 = \mathbf{224}$

Try it: A model shows 3 flats, 11 rods, and 2 units. What number is it?

check: **412**

4. Comparing two models (compare-base-ten-models)

Rule: read both numbers first, then compare **hundreds first**. Only if the hundreds tie do the tens decide; only if the tens tie do the ones decide.
$<$ means *is less than*; $>$ means *is greater than*.

Example: Model A: 2 flats, 5 rods, 1 unit. Model B: 2 flats, 3 rods, 9 units.
Step 1. More single blocks does not mean a bigger number, so I read each model and then compare one place at a time from the left.
Step 2. $\text{A} = 200 + 50 + 1 = 251$, $\text{B} = 200 + 30 + 9 = 239$; hundreds tie, and 5 tens beat 3 tens, so A $>$ B

Try it: Model A: 1 flat, 8 rods, 4 units. Model B: 2 flats, no rods, 3 units.

check: A $<$ B

(!) Watch out: an empty group needs a 0. Three flats and six units is 306, never 36 — the 0 is what keeps the 3 in the hundreds place.